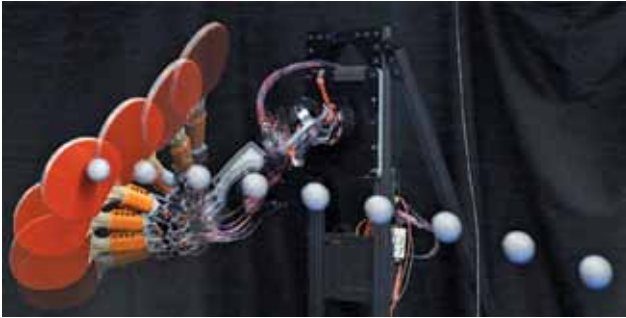




High-speed precision ping pong-playing robot

Engineers at the Massachusetts Institute of Technology (MIT) in Cambridge, Massachusetts Biomimetics Robotics Lab have developed a high-speed ping pong-playing robot featuring a lightweight, multijointed arm, motion-capture

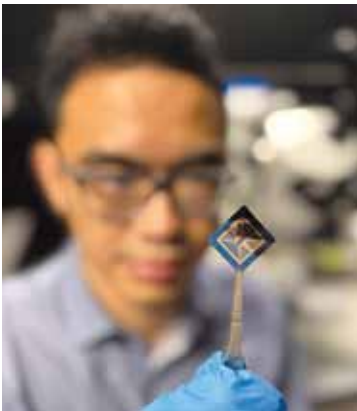


Time lapse photos show a new ping-pong-playing robot performing a top spin. The robot quickly estimates the speed and trajectory of an incoming ball and precisely hits it to a desired location on the table (Credit: Courtesy of the researchers)

cameras, and predictive algorithms. This system benefits from 88% shot return accuracy, reacting with near-human speed and precision. Its advantage lies in adaptability—not just for gameplay but as a platform for advancing humanoid robotics. The research aims to transfer these reactive skills to tasks like search and rescue, pushing boundaries in robotic intelligence and agility.

Brain-inspired device tracks hand movement

At RMIT University (Melbourne, Australia), engineers developed a brain-inspired sensor using molybdenum disulfide (MoS₂) that mimics how the brain sees, processes, and remembers motion in real time. This neuromorphic device processes visual data without the need for external



Brain-inspired device tracks hand movement (Credit: <https://bioengineer.org>)

computing, thereby reducing energy consumption and data load. It offers faster, more efficient performance that is ideal for autonomous systems. Benefits include enhanced motion detection and memory storage. Applications range from safer autonomous vehicles to energy-efficient environmental sensors.

Printable technique for ultrabroadband, sensors adaptable to flexible substrates

Researchers at Chuo University (Hachioji, Japan) led by Assistant Professor Kou Li, developed a fully printable technique for multifunctional image sensor sheets, solving spatial misalignment issues in photo-thermoelectric sensor fabrication. Using carbon nanotube-based inks, their method prints all components on a single platform, ensuring high-yield production and precise alignment. This produces ultrabroadband, sensitive sensors that are adaptable to flexible substrates, benefiting non-destructive inspection.

System for real-time remote control of humanoid robots

Researchers at Stanford University (California) and Simon Fraser University (Burnaby, Canada) developed TWIST, a teleoperation system that enables real-time remote control



The teleoperated whole-body imitation system (TWIST) teleoperates humanoid robots with real-time whole-body human data and AI. It achieves versatile, coordinated, whole-body skills that were not present before (Credit: Yanjie Ze)

of humanoid robots. Tested with the G1 robot, TWIST allows accurate imitation of full-body human motion, including legs, elbows, and waist. This boosts coordination for complex tasks like those in messy kitchens. Benefits include large-scale data collection and adaptability across robot models. Advantages include safer task execution in hazardous areas and future autonomous robot learning.

Glasses that turn speech into text and sign language

Two students from Strawberry Fields High School, Chandigarh earned a design patent (Class 16-06) for GestureX, a wearable device aiding the deaf and hard of hearing. GestureX, a prototype pair of glasses, uses a Raspberry Pi microprocessor, microphone, and Python-based speech recognition to convert speech into text, then displays corresponding sign language videos on an OLED screen. It currently supports ASL, BSL, and ISL.